Printed Pages:02 **Sub Code: KAS101T**

Paper Id: 233026 **Roll No.** | | | | | | | | | | |

B. TECH.
(SEM I) THEORY EXAMINATION 2022-23
ENGINEERING PHYSICS

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

(a) What are the negative results of the Michelson-Morley experiment?
(b) What are the massless particles?
(c) What do you understand by displacement current?
(d) Define the Poynting vector.
(e) What do you understand by a black body?
(f) State Wein's displacement law.
(g) What are the coherent sources?
(h) What is Rayleigh's criterion of resolution?
(i) Define population inversion.
(j) What is the dispersion in optical fibers?

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SECTION B

2. Attempt any *three* of the following: **10x3=30**

(a) Show that momentum of a particle (p) of rest mass (m_o) and energy (E) is given by $E^2 = m_o^2c^4 + p^2c^2$.
(b) If the magnitude of H in a plane wave is 1 amp/meter, find the magnitude of E for a plane wave in free space. (Where $\mu_0 = 4\pi \times 10^{-7}$ Weber/amp-m and $\varepsilon_0 = 8.85 \times 10^{-12}$ C/N-m^2).
(c) Calculate the de-Broglie wavelength associated with a proton moving with a velocity equal to $1/20^{th}$ of the velocity of light.
(d) Calculate the wavelength of light whose first diffraction maximum in the diffraction pattern due to a single slit falls at 30° and coincides with the first minimum of red light of wavelength 6500Å.
(e) Calculate the numerical aperture, acceptance angle, and critical angle of the fiber from the following data: μ_1 (core refractive index) = 1.50 and μ_2 (cladding refractive index) =1.45.

SECTION C

3. Attempt any *one* part of the following: **10x1=10**

(a) Derive the Lorentz Transformation equations and show that Lorentz Transformation approaches to Galilean Transformation when the velocity (v) << c.
(b) Derive a suitable expression for time dilation and show that time dilation is a real effect.

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4. Attempt any *one* part of the following: 10x1=10

(a) Derive the expression for the Poynting theorem in electromagnetic waves.

(b) Show that electromagnetic waves are transverse in nature.

5. Attempt any *one* part of the following: 10x1=10

(a) Derive the time-independent Schrodinger equation, for matter waves.

(b) Derive a suitable expression for Planck's radiation law.

6. Attempt any *one* part of the following: 10x1=10

(a) What are Newton's rings? How they are formed? Derive the expressions for the diameter of bright and dark rings in reflected monochromatic light.

(b) Derive the conditions of principal maxima and minima for the diffraction due to a grating.

7. Attempt any *one* part of the following: 10x1=10

(a) What is an optical fiber? Derive the expression for the numerical aperture, acceptance angle, and critical angle of an optical fiber.

(b) Discuss the construction and working of the He-Ne laser and give its advantages over the Ruby laser.

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